Statistical Analysis Report: Education and Employment Across U.S. Counties

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Conduct a Statistical Analysis Using Python

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Overview

This project examines whether **county-level educational attainment** is associated with **county-level unemployment** in the United States. The **dataset** is the American Community Survey (ACS) 5-Year Estimates (2024), county geography, from the U.S. Census Bureau (data.census.gov); tables B15003 (Educational Attainment) and B23025 (Employment Status). The analysis is entirely statistical: descriptive statistics, visualizations, and two hypothesis tests—a Pearson correlation (education vs. unemployment) and a one-way ANOVA (unemployment rate across Census regions). It does not involve training machine learning models; the goal is to build a clear, reproducible workflow and to interpret results with appropriate attention to limitations and bias (Lusa et al., 2024; Peng, 2011; Sandve et al., 2013).

The main research questions are: (1) *Is the percentage of adults with a bachelor's degree or higher in a county significantly correlated with that county's unemployment rate?* (2) *Does mean unemployment rate differ significantly across U.S. Census regions?* We find a statistically significant negative correlation between education and unemployment, and significant differences in mean unemployment across regions. This report describes the dataset, methods, results, and limitations so that both technical and non-technical readers can understand the findings. Interpretations and conclusions below are grounded in the results and figures produced in the accompanying notebook (*analysis.ipynb*).

Dataset Description

- **Source:** U.S. Census Bureau, American Community Survey (ACS) 5-Year Estimates (2024 release), county geography.
- **Tables:**

  - **B15003** — Educational Attainment for the Population 25 Years and Over (counts by education level).
  - **B23025** — Employment Status for the Population 16 Years and Over (labor force, employed, unemployed, etc.).
- **Unit of analysis:** U.S. counties (and county-equivalents). After merging and dropping rows with missing or suppressed values, the analysis uses over 3,100 counties.
- **Derived variables:**
  - **Percent bachelor's or higher:** (Bachelor's + Master's + Professional + Doctorate) / Total population 25+ × 100, from B15003.
  - **Unemployment rate:** Unemployed / Civilian labor force × 100, from B23025.
- **Categorical variable:** U.S. Census region (Northeast, Midwest, South, West), derived from the state FIPS code in the Census GEO_ID.

The Census Bureau suppresses estimates for small populations or when margins of error are large; those cells appear as "-" or "*****" in the CSV and were treated as missing and excluded from analysis.

Methods

1. **Data loading and cleaning:** B15003 and B23025 CSVs were loaded (skipping the second header row). Numeric columns were coerced to numeric type so suppressed values became missing. The two tables were merged on county GEO_ID. Education and employment derived variables were computed as above. Counties with missing values for either derived variable or for region were dropped. Counties with out-of-range rates (e.g., unemployment < 0 or > 100%) were also dropped.

2. **Descriptive statistics:** Following best practices for initial data analysis (Lusa et al., 2024), summary statistics (count, mean, standard deviation, min, quartiles, max) were computed for percent bachelor's or higher and for unemployment rate. A **summary by region** (count of counties and mean education and mean unemployment per region) was produced to confirm sufficient sample size in each region for the ANOVA and to preview regional differences; a Pearson correlation matrix was also produced to screen the data and guide the choice of hypothesis tests.
3. **Visualizations:** (1) Histograms of percent bachelor's+ and unemployment rate; (2) Scatter plot of percent bachelor's+ vs. unemployment rate; (3) Boxplot of unemployment rate by Census region. All figures have clear titles and labeled axes.
4. **Hypothesis test 1 — Pearson correlation:** A **Pearson correlation test** was used to test for a linear association between percent bachelor's or higher and unemployment rate. The Pearson test is appropriate for two continuous variables when we are interested in linear association and the sample size is large (here, thousands of counties), which supports the normality assumption of the test statistic (e.g., Field, 2018).
    - **H_0:** There is no linear correlation in the population ($\rho = 0$). **H_a:** There is a linear correlation ($\rho \neq 0$). $\alpha = 0.05$ (two-tailed).
5. **Hypothesis test 2 — One-way ANOVA:** A **one-way ANOVA** was used to test whether mean unemployment rate differs across U.S. Census regions (Northeast, Midwest, South, West). ANOVA is appropriate for a continuous outcome and one categorical factor with multiple groups; large sample sizes per region support the assumption that the F-statistic follows the null distribution (Field, 2018).

- **H_0:** Mean unemployment rate is equal across all four regions. **H_a:** At least one region has a different mean. $\alpha = 0.05$.

Results

- **Descriptive statistics:** Mean percent with bachelor's degree or higher was approximately 25%; mean county unemployment rate was approximately 4.7%. Distributions were right-skewed for education and unemployment. The summary by region showed sufficient counties in each region for the ANOVA and previewed how mean education and mean unemployment differ across regions (e.g., which regions tend to have higher or lower averages).
- **Correlation:** The Pearson correlation between percent bachelor's or higher and unemployment rate was **negative** ($r \approx -0.19$). The **p-value** was well below 0.05 (on the order of 10^{-25}), so we **reject the null hypothesis** at $\alpha = 0.05$ and conclude that there is a statistically significant linear association: higher county-level education attainment is associated with lower county-level unemployment.
- **ANOVA:** The one-way ANOVA of unemployment rate by Census region yielded a significant **F-statistic** and **p-value** below 0.05, so we reject the null hypothesis that mean unemployment is equal across regions. At least one pair of regions differs significantly in mean unemployment; the boxplot by region illustrates these differences.
- **Visualizations:** The scatter plot shows a negative trend (more education, lower unemployment). The boxplots by region show variation in unemployment across regions and align with the significant ANOVA result.

Interpretation for a Non-Technical Audience

In plain language: **counties where a larger share of adults have at least a bachelor's degree tend to have lower unemployment rates**, and **average unemployment varies meaningfully by region** (Northeast, Midwest, South, West). The statistical tests indicate that these patterns are unlikely to be due to chance alone. They do **not** mean that getting a degree automatically lowers your personal risk of unemployment, or that any one region is "better"; we are comparing whole counties and regions, not individuals. Policy and resource decisions that affect education and jobs at the county or regional level can use this kind of evidence as one input, along with local context and equity considerations.

Limitations and Potential Bias

- **Dataset limitation:** ACS 5-Year estimates have **margins of error**; small or rural counties often have larger relative uncertainty. We did not incorporate margins of error into the correlation test or visualizations, so precision of estimates varies by county (U.S. Census Bureau, n.d.).
- **Potential bias: Selection bias** may arise from dropping counties with missing or suppressed data. Suppression is more common for small populations, so rural or less populous counties may be under-represented in the analysis. In addition, **ecological fallacy** applies: a county-level association between education and unemployment does not imply the same relationship at the individual level (Robinson, 1950; cited in many statistics and epidemiology texts).
- **Ethical consideration:** Results could be misused to stigmatize low-education counties or to justify reducing investment in areas with higher unemployment. The analysis describes aggregate patterns only and should not be used to infer causes or to make decisions about

individuals or communities without considering equity, structural barriers, and local context.

Next steps: Possible extensions include post-hoc pairwise comparisons (e.g., Tukey) to identify which regions differ; incorporating ACS margins of error into uncertainty quantification; and replicating the analysis with other ACS tables (e.g., industry composition) or other years.

References

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